

L1.6 Trig Derivatives & the Chain Rule

§2.4-2.5 — what you need to know

The six trig derivatives

LEARN THESE THREE

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

GENERATE THESE THREE

$$\frac{d}{dx}(\cos x) = -\sin x$$

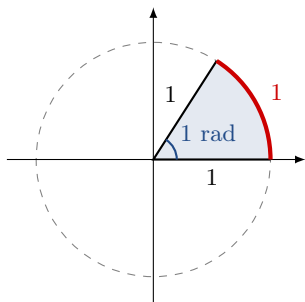
$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

“the derivative of a co-function is always negative”

Look up the function’s derivative on the left. The co-function’s derivative puts *co-* on every function and attaches a minus sign: $\sec x \tan x \rightarrow -\csc x \cot x$.

RADIANS AND DEGREES



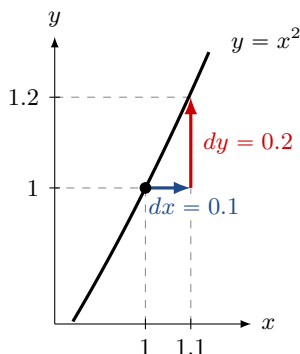
For $y = \sin x$, x is an **angle** and y is a **length**, so $\frac{dy}{dx}$ divides a length by an angle.

Radians put them on even footing. On a unit circle ($r = 1$), an angle of 1 radian subtends an arc of length 1: **angle = length**, in this system only. That is why $\frac{d}{dx} \sin x = \cos x$ holds in radians.

In degrees every trig derivative picks up $\frac{\pi}{180}$: $\frac{d}{dx} \sin(x^\circ) = \frac{\pi}{180} \cos(x^\circ)$.

Differentiating a trig function: radians. Reading a graph or evaluating $\sin 30^\circ$: either unit.

DIFFERENTIALS



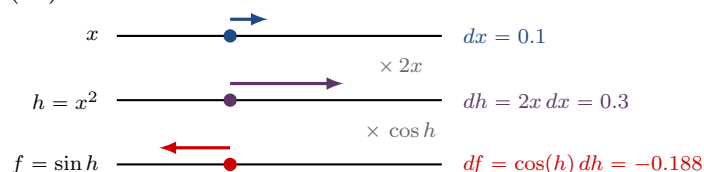
dx is a nudge in the input; dy is a nudge in the output. $\frac{dy}{dx}$ is a ratio of the two, not one symbol.

$$\frac{dy}{dx} = f'(x) \implies dy = f'(x) dx$$

For $y = x^2$, $dy = 2x dx$. At $x = 1$ with $dx = 0.1$:
 $dy = 2(1)(0.1) = 0.2$

THE CHAIN RULE

Nudge the input and watch the nudge come out the far end **changed by a factor at each stage**. For $y = \sin(x^2)$ at $x = 1.5$ with $dx = 0.1$:



Substitute the middle line into the bottom one — $df = \cos(x^2) \cdot 2x dx$ — and divide by dx :

$$\frac{df}{dx} = \cos(x^2) \cdot 2x = f'(h(x)) \cdot h'(x)$$

f' is **evaluated at the inside**, times the derivative of the inside.

COMBINING DERIVATIVE RULES

When a function needs more than one rule, work **from the outside in**. The **outermost operation** is the last step you would take to evaluate the function at a number. Its rule splits the job into pieces, and each piece needs its own rule: $\frac{d}{dx}[fg] = f'g + fg'$ asks for f' and g' separately.

function	outermost	rule for each piece
$(2x + 1)^2 x^3$	product	$(2x + 1)^2$: chain with power x^3 : power
$\frac{\sin x}{x^2 + 1}$	quotient	$\sin x$: trig $x^2 + 1$: power
$\sin(x^2 + 1)$	sine	inside $x^2 + 1$: power
$(x^2 \sin x)^4$	fourth power	inside $x^2 \sin x$: product (x^2 power, $\sin x$ trig)

The chain rule is not a separate choice. Whenever a rule is applied to something other than x , multiply by the derivative of that inside.

WATCH OUT

✗ $\frac{d}{dx}(\sin x^2) = \cos x^2 \rightarrow$ **the inner derivative is missing — it is $\cos(x^2) \cdot 2x$**

✗ $\frac{d}{dx}(\cos x) = \sin x \rightarrow$ **co-functions carry the minus: $-\sin x$**

✗ $\frac{d}{dx}(\sin^2 x) = 2 \sin x \rightarrow$ **the outer function is *squaring*: $2 \sin x \cos x$**

✗ $\frac{dy}{dx}$ is one symbol $\rightarrow$ **it is a ratio of two nudges**

✗ the chain rule multiplies by the derivative of the inside $\rightarrow$ **it is f' *evaluated at the inside*, times the derivative of the inside**

✗ a composite needs one rule $\rightarrow (2x + 1)^2 x^3$: **product, chain on one piece**

✗ $(x^2 \sin x)^4$ starts with the chain rule $\rightarrow$ **it is a fourth *power*; the inside needs product**